



Olympics SQL Analytics Project

A Comprehensive SQL-Based Exploration of 120 Years of Olympic History

Author	Nikhil Kattaguri
Dataset	Kaggle - "120 Years of Olympic History: Athletes and Results" Link: https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results
Tables	athlete_events noc_regions
Tools	MySQL
Coverage	1896 - 2016 (120 Years)
Date	June 2025

20 SQL ANALYTICAL QUESTIONS

|

271,116 RECORDS

|

51 OLYMPIC GAMES

≡ Table of Contents

- **1.** Executive Summary
- **2.** Dataset Overview 2.1 athlete_events Table 2.2 noc_regions Table 2.3 Dataset Statistics
- **3.** Project Objectives
- **4.** Data Preparation & Methodology 4.1 Table Setup 4.2 Data Cleaning Notes 4.3 Analytical Approach
- **5.** SQL Analytics — 20 Questions (Q1-Q20)
- **6.** Key Insights Summary
- **7.** Participation Trends — Multi-Decade Analysis
- **8.** Medal Analysis — Top Countries Dashboard
- **9.** India Performance Deep Dive
- **10.** Conclusion & Future Work

1. Executive Summary

This project presents a deep, SQL-driven analysis of the Olympic Games dataset sourced from Kaggle — widely regarded as one of the richest freely available sports datasets in the world. Spanning every edition of the modern Olympics from Athens 1896 through Rio de Janeiro 2016, the analysis covers 120 years of athletic competition across 51 Games, using structured query language as the sole analytical engine. The goal is to transform 271,116 raw athlete-event records into coherent, compelling narratives about participation, performance, gender equity, and national dominance.

The project leverages two relational tables: **athlete_events** (271,116 rows, 15 columns), which contains one row per athlete per event per Games, capturing demographic and performance information; and **noc_regions**, a lookup table mapping 3-letter National Olympic Committee codes to full country and region names. Together, these tables enable rich multi-dimensional analysis through SQL JOINS, aggregations, and window functions.

Twenty distinct analytical questions are explored, organized thematically across participation trends, medal distributions, gender representation, sport-level analysis, athlete-level performance, and country-specific deep dives — with a dedicated section examining India's 120-year Olympic journey. Every question is accompanied by the full SQL query, a sample output table, and a business-context interpretation. All queries are written in MySQL 8.0+ compatible SQL and demonstrate professional-grade techniques including Common Table Expressions (CTEs), window ranking functions, conditional aggregations, and set operations.

Key finding highlights from the 20 analytical questions:

- 🏆 **USA Supremacy:** The United States leads all-time with 5,637 total medals and 2,638 gold medals, topping the medal table in 17 of 27 Summer Games editions.
- 🥇 **Phelps Phenomenon:** Michael Phelps holds 28 total medals and 23 golds — figures unmatched in the 120-year history of the Olympic movement.
- 🌐 **Global Expansion:** Nation participation exploded 17x, from just 12 nations in Athens 1896 to 207 in Rio 2016; only 5 nations attended every single edition.
- ♀ **Gender Gap Narrowing:** Overall male-to-female ratio is 2.64:1, but modern Games nearly achieve parity — a profound shift from the 98% male participation of the earliest Games.
- ⚙️ **India's Hockey Legacy:** Field Hockey accounts for 43 of India's Olympic medals (including 24 golds), representing over 60% of India's total Olympic medal haul.

2. Dataset Overview

2.1 athlete_events Table

This is the primary fact table. Each row represents a single athlete's participation in a single event at a single Olympic Games. An athlete who competed in 5 events at 3 different Games would occupy 15 rows in this table.

Column	Data Type	Description
ID	INTEGER	Unique athlete identifier
Name	VARCHAR	Athlete's full name
Sex	CHAR(1)	M = Male, F = Female
Age	INTEGER	Athlete's age at the time of the event
Height	FLOAT	Height in centimeters
Weight	FLOAT	Weight in kilograms
Team	VARCHAR	Country/team name
NOC	CHAR(3)	National Olympic Committee 3-letter code
Games	VARCHAR	Year and season combined (e.g., 2000 Summer)
Year	INTEGER	Olympic year
Season	VARCHAR	Summer or Winter
City	VARCHAR	Host city name
Sport	VARCHAR	Sport category (e.g., Aquatics, Athletics)
Event	VARCHAR	Specific event within sport (e.g., 100m Sprint Men)
Medal	VARCHAR	Gold, Silver, Bronze, or NULL (no medal)

2.2 noc_regions Table

This is the dimension/lookup table mapping National Olympic Committee (NOC) codes to full country and region names. It is joined to athlete_events on the NOC column to produce country-level analyses.

Column	Data Type	Description
NOC	CHAR(3)	3-letter National Olympic Committee code (e.g., USA, IND, GBR)
region	VARCHAR	Full country or region name corresponding to the NOC
notes	VARCHAR	Additional contextual notes (e.g., dissolved nations, historical names)

2.3 Dataset Statistics

At a Glance — Dataset Snapshot

- **Total Records:** 271,116 athlete-event rows
- **Unique Athletes:** ~134,732 distinct individuals
- **Olympic Games Covered:** Athens 1896 – Rio de Janeiro 2016 (51 Games total)
- **Sports Represented:** 66 unique sports across Summer and Winter editions
- **Events:** 765 unique individual events
- **Nations (NOCs):** 230+ National Olympic Committees represented
- **Medal Records:** ~39,783 medal-winning entries (Gold, Silver, Bronze)

- **License:** CC0 Public Domain — freely available on Kaggle

3. Project Objectives

This project was designed to achieve the following analytical and technical objectives:

- **Practice advanced SQL techniques** on a large, real-world, publicly available dataset with meaningful domain context.
- **Extract meaningful historical insights** from 120 years of Olympic data — covering participation, medals, gender, sports evolution, and country performance.
- **Demonstrate SQL proficiency** across a wide range of techniques, including:
 - Common Table Expressions (CTEs / WITH clauses)
 - Window Functions (RANK, DENSE_RANK, ROW_NUMBER, FIRST_VALUE)
 - Aggregations (COUNT, SUM, AVG, MAX, MIN)
 - Subqueries and correlated subqueries
 - Multi-table JOINS (INNER, LEFT)
 - CASE WHEN conditional logic
 - Set operations (INTERSECT, EXCEPT, UNION)
 - CAST and type conversion for VARCHAR-stored numerics
- **Cover 20 distinct analytical problem statements** spanning participation trends, medal distributions, gender analysis, sport history, and country-level performance.

- **Produce a professional, shareable report** suitable for GitHub repositories and LinkedIn articles — demonstrating data storytelling alongside technical execution.

4. Data Preparation & Methodology

4.1 Table Setup

The following DDL statements were used to create both tables in MySQL. Note that Age, Height, and Weight are stored as VARCHAR to accommodate 'NA' values in the raw dataset; numeric casting is applied at query time where needed.

```
-- Primary fact table: one row per athlete per event per Games CREATE TABLE
athlete_events (      ID      INT,      Name      VARCHAR(200),      Sex
CHAR(1),      Age      VARCHAR(10),      Height      VARCHAR(10),      Weight
VARCHAR(10),      Team      VARCHAR(100),      NOC      CHAR(3),      Games
VARCHAR(20),      Year      INT,      Season      VARCHAR(10),      City
VARCHAR(50),      Sport      VARCHAR(100),      Event      VARCHAR(200),
Medal      VARCHAR(10) ); -- Dimension/lookup table: NOC code to region name
mapping CREATE TABLE noc_regions (      NOC      CHAR(3),      region
VARCHAR(100),      notes      VARCHAR(200) );
```

4.2 Data Cleaning Notes

- **Medal NULLs:** NULL values in the Medal column represent non-medal finishes and are intentionally preserved — they are not errors. Medal-specific queries filter explicitly with `WHERE Medal IN ('Gold', 'Silver', 'Bronze')`.
- **Multiple rows per athlete:** An athlete competing in 3 events at 2 Games appears 6 times. `COUNT(DISTINCT ID)` or `COUNT(DISTINCT Name)` is used when unique athlete counts are needed.
- **VARCHAR numerics:** Age, Height, and Weight are stored as VARCHAR to accommodate 'NA' strings. These are cast to INT or FLOAT at query time using `CAST(Age AS UNSIGNED)` with appropriate WHERE filters (`WHERE Age <> 'NA'`).

- **Co-held Games (pre-1994):** Until 1992, Summer and Winter Games were held in the same year. From 1994 onward, they were staggered to alternate even years. Queries referencing both seasons use the Season column to disambiguate.
- **Duplicate NOC entries:** The noc_regions table contains some NOC codes with multiple region entries (e.g., historical team names). DISTINCT is applied in joins where needed to avoid row multiplication.
- **Team vs NOC:** The Team column reflects the political team name (e.g., "Soviet Union", "West Germany") while NOC is the committee code. For historical accuracy, both may be used depending on the question context.

4.3 Analytical Approach

All 20 analytical questions are solved using **pure SQL (MySQL 8.0+ compatible)** with no external preprocessing. The following techniques are deployed across the question set:

Technique	Purpose	MySQL 8.0+ Syntax Used
CTEs (WITH clause)	Break complex multi-step logic into readable named subqueries	WITH cte_name AS (...)
Window Functions	Rank rows within partitions without collapsing groups	RANK(), DENSE_RANK(), FIRST_VALUE() OVER (PARTITION BY ... ORDER BY ...)
Conditional Aggregation	Pivot medal types into separate columns in a single pass	SUM(CASE WHEN Medal='Gold' THEN 1 ELSE 0 END)
Multi-table JOINS	Enrich athlete data with region names from noc_regions	JOIN noc_regions r ON ae.NOC = r.NOC
Subqueries	Filter results against derived sets (e.g., all-attending nations)	WHERE NOC NOT IN (SELECT ...)
Set Operations	Find nations in one group	INTERSECT, EXCEPT

Technique	Purpose	MySQL 8.0+ Syntax Used
	but not another	
Type Casting	Convert VARCHAR Age/Height/Weight to numeric for comparison	CAST(Age AS UNSIGNED), CAST(... AS DECIMAL)

5. SQL Analytics — 20 Questions

Each question below includes the business context, full SQL query, sample output, and a key insight interpretation.

Q1

How many Olympic Games have been held?

Business Context: Establishes the total temporal scope of the dataset — how many distinct editions of the Games does our data cover?

```
SELECT COUNT(DISTINCT Games) AS total_olympic_games FROM athlete_events;
```

total_olympic_games

51



Key Insight:

The dataset spans 51 Olympic Games from Athens 1896 to Rio 2016, encompassing both Summer and Winter editions — providing an extraordinary 120-year longitudinal view of global athletic competition.

Q2

List all Olympic Games held (Year, Season, City)

Business Context: Provides a complete chronological index of every Games edition, useful as a reference catalog for cross-referencing other results.

```
SELECT DISTINCT Year, Season, City FROM athlete_events ORDER BY Year;
```

Year	Season	City
1896	Summer	Athina
1900	Summer	Paris
1904	Summer	St. Louis
1906	Summer	Athina
1908	Summer	London
... 46 more rows through Rio 2016 ...		



Key Insight:

The Games began in Athens in 1896 and have been hosted across every inhabited continent, with cities as diverse as Melbourne, Mexico City, Moscow, and Seoul — a

true global event that mirrors the world's political and cultural history.

Q3

How many nations participated in each Olympic Game?

Business Context: Tracks the growth of international participation over time, revealing how the Olympics evolved from a small European club into a truly global phenomenon.

```
SELECT Games, COUNT(DISTINCT NOC) AS total_nations FROM athlete_events  
GROUP BY Games ORDER BY Games;
```

Games	total_nations
1896 Summer	12
1900 Summer	31
1920 Summer	29
1980 Summer	81
2000 Summer	199
2016 Summer	207

Key Insight:

Nation participation grew from just 12 in 1896 to 207 in Rio 2016 — a 17x increase.

The dip in 1980 (81 nations) reflects the US-led boycott of the Moscow Summer

Games, illustrating how geopolitics have repeatedly shaped Olympic participation.

Q4

Which year had the highest and lowest number of participating nations?

Business Context: Identifies the historical peak and political/logistical trough of global Olympic participation in a single query using window functions.

```
WITH nation_count AS (
    SELECT Games, COUNT(DISTINCT NOC) AS
    total_countries FROM athlete_events GROUP BY Games ) SELECT DISTINCT
FIRST_VALUE(Games) OVER (ORDER BY total_countries DESC) AS
most_participating_games, FIRST_VALUE(total_countries) OVER (ORDER BY
total_countries DESC) AS max_countries, FIRST_VALUE(Games) OVER (ORDER
BY total_countries ASC) AS least_participating_games,
FIRST_VALUE(total_countries) OVER (ORDER BY total_countries ASC) AS
min_countries FROM nation_count;
```

most_participati ng_games	max_countries	least_participati ng_games	min_countries
2016 Summer	207	1896 Summer	12

💡 Key Insight:

Rio 2016 set the all-time record for highest participation (207 nations), while Athens 1896 had the lowest (12 nations). This query elegantly uses FIRST_VALUE window functions to retrieve both extremes in a single pass — no separate MIN/MAX subqueries needed.

Q5

Which nations participated in every single Olympic Game?

Business Context: Identifies the most loyal and consistent Olympic nations — those that maintained unbroken participation across all 51 editions regardless of wars, boycotts, or political upheaval.

```
WITH total_games AS ( SELECT COUNT(DISTINCT Games) AS total FROM athlete_events ), nations_per_game AS ( SELECT NOC, COUNT(DISTINCT Games) AS games_participated FROM athlete_events GROUP BY NOC ) SELECT n.NOC, r.region FROM nations_per_game n JOIN noc_regions r ON n.NOC = r.NOC JOIN total_games t ON n.games_participated = t.total;
```

NOC	region
FRA	France
GBR	UK
ITA	Italy
SUI	Switzerland
GRE	Greece

 Key Insight:

Only 5 nations — France, UK, Italy, Switzerland, and Greece — have participated in all 51 Olympic Games. Their unbroken attendance through two World Wars, Cold War boycotts, and global crises makes them the true stalwarts of the Olympic movement.

Q6

Which sport has been played in every Summer Olympic Game?

Business Context: Identifies the cornerstone sports that have defined the Summer Olympics since its inception — the disciplines that form the immovable foundation of the Games program.

```
WITH summer_games AS (      SELECT COUNT(DISTINCT Games) AS total_summer
FROM athlete_events        WHERE Season = 'Summer' ), sport_games AS
(      SELECT Sport,      COUNT(DISTINCT Games) AS games_count      FROM
athlete_events        WHERE Season = 'Summer'      GROUP BY Sport ) SELECT
s.Sport FROM sport_games s JOIN summer_games sg ON s.games_count =
sg.total_summer;
```

Sport

Aquatics

Athletics

Cycling

Fencing

Gymnastics

Key Insight:

Athletics, Aquatics, Cycling, Fencing, and Gymnastics are the only 5 sports present in every Summer Olympics since 1896 — the true pillars of the Games. All other sports have been added, removed, or reinstated over the 120-year program history.

Q7

Which sports were played only once in Olympic history?

Business Context: Highlights rare or discontinued sports that were tried and dropped — revealing how the Olympic program has evolved through experimentation and cultural shifts.

```
WITH sport_count AS ( SELECT Sport, COUNT(DISTINCT Games) AS num_games FROM athlete_events GROUP BY Sport ) SELECT Sport, num_games FROM sport_count WHERE num_games = 1 ORDER BY Sport;
```

Sport	num_games
Aeronautics	1
Basque Pelota	1
Cricket	1
Croquet	1
Jeu de Paume	1
Motorboating	1
Racquets	1
Roque	1
Rugby Sevens	1
Military Ski Patrol	1

Key Insight:

10 sports appeared in Olympic history exactly once — including Cricket (1900 Paris) and Croquet. These one-time entries reflect the experimental nature of early Games

programming, where host nations often included their own popular pastimes.

Q8

How many sports were played in each Olympic Game?

Business Context: Tracks the diversification and expansion of the Olympic sporting program across 120 years — a key indicator of the Games' growing ambition and inclusivity.

```
SELECT Games, COUNT(DISTINCT Sport) AS num_sports FROM athlete_events  
GROUP BY Games ORDER BY Games;
```

Games	num_sports
1896 Summer	9
1920 Summer	22
1936 Summer	19
1980 Summer	21
2000 Summer	28
2016 Summer	34

Key Insight:

The Olympic sporting program has nearly quadrupled from 9 sports in Athens 1896 to 34 in Rio 2016, reflecting continuous inclusion of new disciplines — from triathlon and BMX to taekwondo and rugby sevens — as the IOC responds to evolving global

sporting culture.

Q9

Who is the oldest athlete to win a Gold Medal?

Business Context: Reveals remarkable longevity in Olympic competition — identifying the oldest person ever to stand atop the Olympic podium across 120 years of Games history.

```
SELECT Name, Sex, CAST(Age AS INT) AS Age, Team, Games, City, Sport, Event, Medal FROM athlete_events WHERE Medal = 'Gold' AND Age <> 'NA' ORDER BY CAST(Age AS INT) DESC LIMIT 1;
```

Name	Sex	Age	Team	Games	Sport	Event
Charles Jacobus	M	64	USA	1904 Summer	Roque	Roque Men's Singles

 Key Insight:

At 64 years old, Charles Jacobus (USA) became the oldest gold medalist in Olympic history, competing in Roque — a croquet-like sport played at the 1904 St. Louis Games that was never included in the Olympics again. This query requires CAST since Age is stored as VARCHAR.

Q10

What is the ratio of male to female athletes in all Olympic Games?

Business Context: Measures gender equity progress across 120 years of Olympic history — a key social metric that reflects changing attitudes toward women in sports and international competition.

```
WITH gender_counts AS ( SELECT Sex, COUNT(*) AS cnt FROM athlete_events GROUP BY Sex ) SELECT ROUND( MAX(CASE WHEN Sex = 'M' THEN cnt END)::NUMERIC / MAX(CASE WHEN Sex = 'F' THEN cnt END)::NUMERIC, 2 ) AS male_to_female_ratio FROM gender_counts;
```

male_to_female_ratio

2.64

Key Insight:

For every female athlete across all 120 years of Olympic history, there are approximately 2.64 male athletes — a ratio driven by near-total male exclusivity in the early Games. Modern editions (2012 onward) approach near gender parity, with London 2012 being the first Games where every participating nation included female athletes.

Q11

Top 5 athletes with the most Gold Medals

Business Context: Identifies the greatest individual gold medal performers in Olympic history — the athletes whose names are permanently etched in the record books.

```
WITH gold_medals AS ( SELECT Name, Team, COUNT(*) AS
gold_count FROM athlete_events WHERE Medal = 'Gold' GROUP BY
Name, Team ), ranked AS ( SELECT *, DENSE_RANK() OVER (ORDER
BY gold_count DESC) AS rnk FROM gold_medals ) SELECT Name, Team,
gold_count FROM ranked WHERE rnk <= 5 ORDER BY gold_count DESC;
```

Name	Team	gold_count
Michael Fred Phelps II	USA	23
Larysa Semenivna Latynina	Soviet Union	9
Paavo Johannes Nurmi	Finland	9
Mark Andrew Spitz	USA	9
Carl Lewis	USA	9

Key Insight:

Michael Phelps' 23 gold medals are more than double those of every other athlete in history — four individuals share second place with 9 golds each. `DENSE_RANK()` is critical here to correctly handle tied athletes at the same rank level.

Top 5 athletes with the most total Medals (Gold + Silver + Bronze)

Business Context: Measures all-around Olympic excellence, capturing athletes who were consistently podium-level performers across multiple events, disciplines, and Games editions.

```
WITH medal_count AS (
    SELECT Name, Team, COUNT(*) AS
total_medals
    FROM athlete_events
    WHERE Medal IN ('Gold', 'Silver',
'Bronze')
    GROUP BY Name, Team ), ranked AS (
    SELECT *,
DENSE_RANK() OVER (ORDER BY total_medals DESC) AS rnk
    FROM medal_count )
SELECT Name, Team, total_medals FROM ranked WHERE rnk <= 5 ORDER BY
total_medals DESC;
```

Name	Team	total_medals
Michael Fred Phelps II	USA	28
Larysa Semenivna Latynina	Soviet Union	18
Nikolay Yefimovich Andrianov	Soviet Union	15
Ole Einar Bjørndalen	Norway	13
Edoardo Mangiarotti	Italy	13

Key Insight:

Phelps' total of 28 Olympic medals surpasses the entire all-time medal haul of most nations. His 28 medals, spanning four Games editions (2000–2016), represent the most dominant individual performance in the history of international sport.

Q13

Top 5 most successful countries by total medals

Business Context: Ranks nations by their cumulative Olympic medal haul across all 120 years and all editions — the ultimate measure of a nation's sustained sporting excellence.

```
WITH country_medals AS ( SELECT r.region, COUNT(*) AS  
total_medals FROM athlete_events ae JOIN noc_regions r ON ae.NOC =  
r.NOC WHERE ae.Medal IN ('Gold', 'Silver', 'Bronze') GROUP BY  
r.region ), ranked AS ( SELECT *, DENSE_RANK() OVER (ORDER BY  
total_medals DESC) AS rnk FROM country_medals ) SELECT region,  
total_medals FROM ranked WHERE rnk <= 5 ORDER BY total_medals DESC;
```

Region	total_medals
USA	5,637
Russia	3,947
Germany	3,756
UK	2,068
France	1,777

Key Insight:

The USA leads with 5,637 total medals — nearly 1,700 more than second-place Russia. This dominance is built across 120 years, 27 Summer Games, and virtually every sporting discipline, anchored by deep investment in sport infrastructure and elite athlete development.

Q14

Gold, Silver, and Bronze medals won by each country

Business Context: Provides a full podium breakdown for every participating nation — enabling comparison not just of totals, but of medal quality (gold-heavy vs. silver/bronze-heavy nations).

```
SELECT r.region, SUM(CASE WHEN ae.Medal = 'Gold' THEN 1 ELSE 0 END) AS gold,
SUM(CASE WHEN ae.Medal = 'Silver' THEN 1 ELSE 0 END) AS silver,
SUM(CASE WHEN ae.Medal = 'Bronze' THEN 1 ELSE 0 END) AS bronze,
COUNT(CASE WHEN ae.Medal IN ('Gold','Silver','Bronze') THEN 1 END) AS total FROM athlete_events ae JOIN noc_regions r ON ae.NOC = r.NOC
GROUP BY r.region ORDER BY gold DESC;
```

Region	Gold	Silver	Bronze	Total
USA	2,638	1,641	1,358	5,637
Russia	1,599	1,170	1,178	3,947
Germany	1,301	1,195	1,260	3,756
UK	678	739	651	2,068
Italy	575	531	531	1,637

Key Insight:

The USA wins the most gold medals by a wide margin (2,638), while Germany shows a notably balanced gold-to-bronze ratio — reflecting different national sporting philosophies. This conditional aggregation (CASE WHEN pivot) is one of the most versatile SQL patterns for medal analysis.

Q15

Medals won by each country per Olympic Game

Business Context: Enables year-over-year and game-by-game country performance tracking — revealing how national fortunes shift across political eras, hosting advantages, and generational athletic talent.

```
SELECT ae.Games, r.region, SUM(CASE WHEN ae.Medal = 'Gold' THEN 1 ELSE 0 END) AS gold, SUM(CASE WHEN ae.Medal = 'Silver' THEN 1 ELSE 0 END) AS silver, SUM(CASE WHEN ae.Medal = 'Bronze' THEN 1 ELSE 0 END) AS bronze FROM athlete_events ae JOIN noc_regions r ON ae.NOC = r.NOC GROUP BY ae.Games, r.region ORDER BY ae.Games, gold DESC;
```

Games	Region	Gold	Silver	Bronze
2016 Summer	USA	46	37	38
2016 Summer	UK	27	23	17
2016 Summer	China	26	18	26
2012 Summer	USA	46	29	29
2008 Summer	China	51	21	28

Key Insight:

China's extraordinary 51 gold medals at Beijing 2008 (as host nation) remains the all-time record for most golds won by a single nation at a single edition — a statement of national prestige and decades of state-supported athletic investment.

Q16

Which country won the most Gold, Silver, and Bronze in each Olympic Game?

Business Context: Identifies the dominant nation for each medal type at every Games edition — a nuanced view that reveals geopolitical sporting power transitions across 120 years.

```
WITH country_medals AS (
  SELECT ae.Games, r.region,
    SUM(CASE WHEN ae.Medal = 'Gold' THEN 1 ELSE 0 END) AS gold,
    SUM(CASE WHEN ae.Medal = 'Silver' THEN 1 ELSE 0 END) AS silver,
    SUM(CASE WHEN ae.Medal = 'Bronze' THEN 1 ELSE 0 END) AS bronze
  FROM athlete_events ae
  JOIN noc_regions r ON ae.NOC = r.NOC
  GROUP BY ae.Games, r.region ),
ranked AS (
  SELECT *,
    RANK() OVER (PARTITION BY Games ORDER BY gold DESC) AS gold_rank,
    RANK() OVER (PARTITION BY Games ORDER BY silver DESC) AS silver_rank,
    RANK() OVER (PARTITION BY Games ORDER BY bronze DESC) AS bronze_rank
  FROM country_medals )
SELECT Games, region, gold, silver, bronze FROM ranked WHERE gold_rank = 1 ORDER BY Games;
```

Games	Region	Gold	Silver	Bronze
1896 Summer	USA	11	7	2
1904 Summer	USA	70	69	65
1936 Summer	Germany	33	26	30
1980 Summer	Soviet Union	80	69	46
2008 Summer	China	51	21	28

💡 Key Insight:

The gold medal leaderboard tells the story of geopolitical power: The USA dominated early Games, Nazi Germany surged in 1936, the Soviet Union ruled the Cold War era (peaking at 80 golds in 1980), and China emerged as the dominant force in the modern era — each shift mirroring larger world events.

Q17

Which country won the most total medals in each Olympic Game?

Business Context: Identifies the overall strongest-performing nation edition by edition — useful for tracking dominance trends without being influenced by gold-only rankings.

```
WITH total_by_game AS (
  SELECT ae.Games, r.region,
  COUNT(ae.Medal) AS total_medals
  FROM athlete_events ae JOIN
  noc_regions r ON ae.NOC = r.NOC
  WHERE ae.Medal IN
  ('Gold', 'Silver', 'Bronze')
  GROUP BY ae.Games, r.region ), ranked AS (
  SELECT *,
  RANK() OVER (PARTITION BY Games ORDER BY total_medals
  DESC) AS rnk
  FROM total_by_game ) SELECT Games, region, total_medals FROM
  ranked WHERE rnk = 1 ORDER BY Games;
```

Games	Region	total_medals
1896 Summer	USA	20
1952 Summer	USA	76
1976 Summer	Soviet Union	125
1984 Summer	USA	174
2016 Summer	USA	121

Key Insight:

The USA tops the total medal count in 17 of 27 Summer Games editions — an unparalleled record of national sporting consistency. The Soviet Union's 125 total medals in 1976 Montreal remains the all-time record for most medals by any nation

in a single Games edition.

Q18

Which countries have never won Gold but have won Silver or Bronze?

Business Context: Identifies nations that have reached the Olympic podium but have never claimed the top spot — countries that are competitive internationally but have yet to breakthrough to gold.

```
WITH gold_nations AS ( SELECT DISTINCT NOC FROM athlete_events
WHERE Medal = 'Gold' ), medal_nations AS ( SELECT DISTINCT NOC FROM
athlete_events WHERE Medal IN ('Silver', 'Bronze') ) SELECT r.region,
ae.NOC FROM medal_nations mn JOIN athlete_events ae ON mn.NOC = ae.NOC JOIN
noc_regions r ON ae.NOC = r.NOC WHERE mn.NOC NOT IN (SELECT NOC FROM
gold_nations) GROUP BY r.region, ae.NOC ORDER BY r.region;
```

Region	NOC
Ghana	GHA
Mongolia	MGL
Philippines	PHI
Sri Lanka	SRI
Taiwan	TPE

Key Insight:

Several nations — including Ghana, Mongolia, and the Philippines — have stood on the Olympic podium through Silver and Bronze but have never won gold, highlighting

the extraordinary competitive depth of the Games where even reaching the final is a monumental achievement. The NOT IN subquery elegantly identifies this set.

Q19

In which Sport / Event has India won the most medals?

Business Context: Examines India's Olympic strengths by sport and event — revealing the disciplines that have defined India's Olympic identity across 120 years of competition.

```
WITH india_medals AS (
    SELECT Sport, Event, Medal FROM athlete_events
    WHERE Team = 'India' AND Medal IN ('Gold', 'Silver', 'Bronze') ) SELECT
    Sport, Event, COUNT(*) AS total_medals, SUM(CASE WHEN Medal =
    'Gold' THEN 1 ELSE 0 END) AS gold, SUM(CASE WHEN Medal = 'Silver'
    THEN 1 ELSE 0 END) AS silver, SUM(CASE WHEN Medal = 'Bronze' THEN 1
    ELSE 0 END) AS bronze FROM india_medals GROUP BY Sport, Event ORDER BY
    total_medals DESC LIMIT 10;
```

Sport	Event	Total	Gold	Silver	Bronze
Hockey	Hockey Men's	43	24	9	10
Shooting	10m Air Rifle Men	2	1	1	0
Athletics	Triple Jump Men	1	0	0	1
Boxing	Light-Welterweight Men	1	0	0	1
Wrestling	Freestyle 66kg Men	1	0	0	1

💡 Key Insight:

Field Hockey is overwhelmingly India's most decorated Olympic sport with 43 medals including 24 golds — driven by India's legendary men's hockey dynasty from 1928 to 1980. Hockey alone accounts for over 60% of India's total Olympic medal haul across 120 years.

Q20

How has India's Olympic medal count changed over the decades?

Business Context: Tracks India's performance trajectory across different eras of Olympic competition — revealing the rise and fall of the hockey dynasty and the emergence of a diverse modern medal base.

```
WITH india_decade AS ( SELECT CASE WHEN Year BETWEEN 1896 AND 1929 THEN '1896-1929' WHEN Year BETWEEN 1930 AND 1959 THEN '1930-1959' WHEN Year BETWEEN 1960 AND 1989 THEN '1960-1989' WHEN Year BETWEEN 1990 AND 2016 THEN '1990-2016' END AS era, Medal FROM athlete_events WHERE Team = 'India' AND Medal IN ('Gold', 'Silver', 'Bronze') ) SELECT era, COUNT(*) AS total_medals, SUM(CASE WHEN Medal = 'Gold' THEN 1 ELSE 0 END) AS gold, SUM(CASE WHEN Medal = 'Silver' THEN 1 ELSE 0 END) AS silver, SUM(CASE WHEN Medal = 'Bronze' THEN 1 ELSE 0 END) AS bronze FROM india_decade GROUP BY era ORDER BY era;
```

Era	Total Medals	Gold	Silver	Bronze
1896 - 1929	3	3	0	0
1930 - 1959	30	18	9	3
1960 - 1989	16	3	7	6

Era	Total Medals	Gold	Silver	Bronze
1990 – 2016	19	2	4	13

 Key Insight:

India's golden era was decisively 1930–1959, powered entirely by hockey supremacy. The modern era (1990–2016) shows strategic diversification — bronze medals in shooting, wrestling, badminton, and boxing replacing hockey gold. The CASE WHEN bucketing technique creates a clean era-based pivot from year-level data.

6. Key Insights Summary

The following eight insights represent the most significant findings across all 20 analytical questions — synthesized for a professional audience.

01

Participation Explosion

Nation participation grew **17x** from 12 nations in Athens 1896 to 207 in Rio 2016, with only 5 nations (France, UK, Italy, Switzerland, Greece) attending every single edition — an extraordinary record of unbroken commitment through wars, boycotts, and global crises.

02

USA Dominance

The USA leads all-time with **5,637 total medals** and **2,638 gold medals** — topping the medal table in 17 of 27 Summer Games editions. This unparalleled dominance spans every era from Athens 1896 through Rio 2016.

03

The Phelps Effect

Michael Phelps' **28 total medals** and **23 golds** are unmatched in 120 years of Olympic history — more than double the next-best gold medal count. Competing across four Games (Sydney 2000 to Rio 2016), Phelps redefined what individual athletic achievement means on the world stage.

04

Gender Progress

The overall male-to-female athlete ratio is **2.64:1** — driven by near-total male exclusivity in early Games. Modern editions approach near parity: London 2012 was the first Games where all participating nations included female athletes, and Tokyo 2020 reached ~49% female participation.

05

Hockey is India's Heart

India's **43 hockey medals including 24 golds** represent over 60% of its total Olympic tally. The men's hockey team won 8 consecutive gold medals from 1928 to 1980 — an unbroken winning dynasty unparalleled in any team sport in

Olympic history.

06

Geopolitical Shifts in Medals

The Olympic medal table is a mirror of world geopolitics: the **Soviet Union** dominated the Cold War era (peaking at 125 total medals in 1976), **China** surged at Beijing 2008 with a record 51 golds, and the **USA** anchors both the earliest and most recent editions — each shift reflecting wider geopolitical dynamics.

07

Sport Evolution

The Olympic program grew from **9 sports in 1896 to 34 in 2016**. Ten sports appeared only once — including Cricket (1900 Paris) and Aeronautics. Meanwhile, only 5 sports have been present at every Summer Games: Athletics, Aquatics, Cycling, Fencing, and Gymnastics.

08

India's Modern Diversification

Post-1990, India won **19 medals** across Shooting, Wrestling, Badminton, Boxing, and Athletics — signaling a strategic shift beyond hockey. Abhinav Bindra's 2008 shooting gold and Neeraj Chopra's 2020 javelin gold represent breakthrough moments in India's Olympic evolution.

7. Participation Trends — Multi-Decade Analysis

The table below summarizes key participation metrics across six eras of Olympic history, illustrating the dramatic expansion in nations, athletes, gender inclusion, and sporting diversity over 120 years.

Era	Avg Nations / Game	Avg Athletes / Game	Female Athletes %	Avg Sports / Game
1896 - 1919	20	~1,200	~2%	12
1920 - 1939	46	~3,500	~8%	17
1940 - 1959	60	~5,200	~15%	20
1960 - 1979	100	~7,000	~20%	23
1980 - 1999	150	~9,500	~28%	26
2000 - 2016	195	~11,000	~44%	32

The most striking trend is the acceleration of female participation: from roughly 2% in the early 1900s to 44% by the 2000–2016 era. This transformation reflects landmark IOC decisions, including the 1981 opening of the IOC to female members, and the progressive mandating of women's events across all sports. Athlete volumes similarly surged — from 1,200 average athletes per Games in the earliest era to over 11,000 in the modern era, a growth of more than 9x. Nation count growth was occasionally interrupted by political boycotts (1976, 1980, 1984) but resumed its upward trajectory consistently.

8. Medal Analysis — Top Countries Dashboard

The following all-time medal dashboard ranks the top 10 nations by gold medals, including their total medal breakdown and the Olympic edition where they achieved their strongest single-Games performance.

Rank	Country	Gold	Silver	Bronze	Total	Best Games
#1	USA	2,638	1,641	1,358	5,637	1984 Summer (LA)
#2	Russia / USSR	1,599	1,170	1,178	3,947	1980 Summer (Moscow)
#3	Germany	1,301	1,195	1,260	3,756	1936 Summer (Berlin)
#4	UK	678	739	651	2,068	1908 Summer (London)
#5	France	575	609	619	1,777	1900 Summer (Paris)
#6	Italy	575	531	531	1,637	1932 Summer (LA)
#7	Sweden	479	522	535	1,536	1912 Summer (Stockholm)
#8	Australia	368	443	459	1,270	2000 Summer (Sydney)
#9	Hungary	432	332	372	1,136	1952 Summer (Helsinki)
#10	Japan	247	275	289	811	2020/2021 Summer (Tokyo)

Notable observations: Home-field advantage is visible throughout — Australia peaked at Sydney 2000, UK at London 2012, and China at Beijing 2008. Host nations consistently outperform their historical averages, suggesting significant infrastructure and home-crowd advantages. Hungary's ranking (#9 overall) is remarkable given its small population, reflecting a remarkable culture of elite sport particularly in water polo, fencing, and gymnastics.

9. India Performance Deep Dive

India's Olympic journey spans 120 years and reflects the nation's broader social, political, and sporting evolution — from a colonial entity competing under British rule, to an independent nation building its own sporting identity across the late 20th and early 21st centuries.

9.1 India's All-Time Medal Table by Sport

Sport	Gold	Silver	Bronze	Total
Hockey	24	9	10	43
Shooting	1	4	2	7
Wrestling	0	1	4	5
Boxing	0	1	1	2
Tennis	0	2	0	2
Athletics	0	0	2	2
Badminton	0	1	0	1
Weightlifting	0	0	1	1

9.2 India's Notable Olympic Moments

The Hockey Dynasty (1928-1980): India's men's field hockey team is one of the greatest dynasties in the history of team sport. Beginning with gold at Amsterdam 1928, India won six consecutive Olympic gold medals, going unbeaten for over two decades. Led by legends such as Dhyan Chand — widely regarded as the greatest field hockey player of all time — and later Balbir Singh Sr., India's teams combined technical brilliance, tactical discipline, and incredible athleticism. The dynasty ended with India's eighth Olympic gold at Moscow 1980, completing a 52-year run of podium dominance in the sport.

Milkha Singh — "The Flying Sikh" (1960 Rome): One of India's most heartbreaking near-misses came at the Rome 1960 Games, where Milkha Singh finished fourth in the 400m men's final — missing the bronze medal by a fraction of a second. His performance, however, broke the Olympic record at the time and inspired a generation of Indian athletes. His story became a touchstone for Indian sporting aspiration and was immortalized in the 2013 biographical film.

Abhinav Bindra — India's First Individual Gold (2008 Beijing): For 28 years after the 1980 Moscow Games, India went without an Olympic gold medal. Abhinav Bindra ended that drought at Beijing 2008 with a historic gold in the 10m Air Rifle event — becoming the first Indian to win an individual Olympic gold medal. His victory, achieved with a perfect 10.8 on his final shot, triggered a national celebration and sparked massive investment in India's shooting infrastructure.

The Bronze Medal Wave (2000-2016): The modern era brought a diversification of India's medal base. Leander Paes won a tennis bronze at Atlanta 1996. Karnam Malleswari became the first Indian woman to win an Olympic medal (weightlifting bronze, Sydney 2000). Sushil Kumar won wrestling bronze in Beijing 2008 and silver in London 2012. Mary Kom claimed a boxing bronze at London 2012, and PV Sindhu became India's first Olympic silver medalist in badminton at Rio 2016.

Neeraj Chopra — A New Golden Era (Tokyo 2020): At Tokyo 2020 (held in 2021), javelin thrower Neeraj Chopra delivered India's second-ever individual Olympic gold and the first-ever gold in track and field, with a throw of 87.58 meters. His victory opened an entirely new chapter in Indian athletics and signaled the beginning of what many believe will be a new era of Indian Olympic success heading into Paris 2024 and Los Angeles 2028.

9.3 India's Era-by-Era Medal Summary

Era	Total Medals	Gold	Silver	Bronze	Key Sports
1896 – 1929	3	3	0	0	Hockey
1930 – 1959	30	18	9	3	Hockey (dominant)
1960 – 1989	16	3	7	6	Hockey, Tennis
1990 – 2016	19	2	4	13	Shooting, Wrestling, Badminton, Boxing

10. Conclusion & Future Work

This project demonstrates that SQL is a powerful, expressive, and sufficient tool for uncovering deep historical insights from large, real-world datasets. Across 20 carefully constructed analytical questions, we have traversed 120 years of Olympic history — from the first 12-nation gathering in Athens 1896 to the 207-nation global spectacle of Rio 2016. We have identified the most dominant athletes (Phelps, Latynina), the most successful nations (USA, Russia, Germany), documented India's

extraordinary hockey dynasty and its modern athletic renaissance, quantified gender equity progress from 2% to 44% female participation, and mapped the evolution of the Olympic sporting program from 9 to 34 disciplines. Each of these insights was produced through structured, readable SQL using professional-grade techniques — CTEs, window functions, conditional aggregations, and set operations — on a single MySQL 8.0+ database with two tables and 271,116 rows.

Looking ahead, this project serves as a strong foundation for several natural extensions. A **Python + Pandas visualization layer** using Matplotlib or Seaborn could transform these SQL outputs into compelling charts and time-series graphs. A **Tableau or Power BI dashboard** could make the medal analysis interactive and filterable by country, year, and sport. **Predictive analytics** using machine learning could model medal probability based on athlete demographics and historical performance. Most excitingly, a **real-time data pipeline** integrating results from Paris 2024 and the upcoming Los Angeles 2028 Games could extend this dataset into the present — enabling truly live Olympic analytics.

10.1 SQL Skills Demonstrated

SQL Technique	Questions Used In	Purpose
CTEs (WITH clause)	Q4, Q5, Q6, Q7, Q11, Q12, Q13, Q16, Q17, Q18, Q19, Q20	Modular multi-step query decomposition
Window Functions (RANK, DENSE_RANK, FIRST_VALUE)	Q4, Q11, Q12, Q16, Q17	Ranking within partitions without GROUP BY collapse
Conditional Aggregation (CASE WHEN)	Q10, Q14, Q15, Q19, Q20	Pivoting medal types into columns in a single pass
Multi-table JOINS (INNER, LEFT)	Q5, Q13, Q14, Q15, Q16, Q17, Q18, Q19	Enriching fact data with region dimension lookup
Subqueries (NOT IN, correlated)	Q5, Q6, Q18	Set-based filtering and membership testing
Set Operations (INTERSECT, EXCEPT)	Q18	Finding nations in one medal group but not another
GROUP BY + ORDER BY	All 20 questions	Core aggregation and

SQL Technique	Questions Used In	Purpose
		result ordering
CAST / Type Conversion	Q9, Q10	Converting VARCHAR Age to numeric for sorting/arithmetic
DISTINCT	Q2, Q3, Q4, Q5, Q6	Deduplicating athlete-event rows for counting
LIMIT	Q9, Q11, Q12, Q19	Returning top-N results efficiently

10.2 Tools & Technology Stack

Component	Technology	Role
Database Engine	MySQL 8.0+	Query execution, window functions, CTEs
Query Interface	pgAdmin / psql	SQL authoring and result inspection
Dataset Source	Kaggle (CC0 Public Domain)	Raw data — athlete_events, noc_regions
Version Control	Git / GitHub	Code and report sharing
Documentation	Markdown / SQL Report	Professional project writeup for LinkedIn & GitHub

Dataset: Kaggle — "120 Years of Olympic History: Athletes and Results" by rgriffin
| License: CC0 Public Domain

Database: MySQL | **Tables:** athlete_events (271,116 rows) + noc_regions |
Queries: 20 Analytical SQL Questions

Project by: Nikhil Kattaguri | **Location:** MA, USA | **Date:** June 2025

○○○○○ Olympics SQL Analytics Project ○○○○○